# Nipple-sparing Mastectomy: What the Radiologist Should Know

*Samantha C. Lee, MBBS*
*Karen Mendez-Broomberg, MD, MPH*
*Katherine Eacobacci, MSIV*
*Nina S. Vincoff, MD*
*Ekta Gupta, MD*
*Suzanne E. McElligott, MD*

Nipple-sparing mastectomy (NSM) is increasingly offered to patients undergoing treatment of breast cancer and prophylaxis treatment for reduction of breast cancer risk. NSM is considered oncologically safe for appropriately selected patients and is associated with improved cosmetic outcomes and quality of life. Accepted indications for NSM have expanded in recent years, and currently only inflammatory breast cancer or malignancy involving the nipple is considered an absolute contraindication. Neoplasms close to and involving the nipple areolar complex are common, and cancer of the lactiferous ducts can spread to the nipple. Therefore, accurate determination of nipple involvement at imaging examinations is critical to identifying appropriate candidates for NSM and preventing local recurrence. Multiple imaging features have been described as predictors of nipple involvement, with tumor to nipple distance, enhancement between the index malignancy and the nipple, and nipple retraction demonstrating the highest predictive values. These features can be assessed at multimodality breast imaging, particularly at breast MRI, which demonstrates high specificity and negative predictive value for determining nipple involvement in malignancy.

*Online supplemental material is available for this article.*



**Abbreviations:** ACR = American College of Radiology, BI-RADS = Breast Imaging Reporting and Data System, DCIS = ductal carcinoma in situ, ER = estrogen receptor, HER2 = human epidermal growth factor receptor 2, IDC = invasive ductal carcinoma, NAC = nipple areolar complex, NSM = nipple-sparing mastectomy, PR = progesterone receptor, TND = tumor-to-nipple distance

**RadioGraphics 2022; 42:321–339**

**https://doi.org/10.1148/rg.210136**

**Content Code:** BR

From the Department of Radiology, Donald and Barbara Zucker School of Medicine at Hofstra University/Northwell Health System, 300 Community Dr, Manhasset, NY 11030. Presented as an education exhibit at the 2020 RSNA Annual Meeting. Received April 19, 2021; revision requested May 25 and received July 20; accepted July 26. For this journal-based SA-CME activity, the authors, editor, and reviewers have disclosed no relevant relationships. **Address correspondence to** S.L. (e-mail: *sclee0414@gmail.com*).



## SA-CME LEARNING OBJECTIVES

*After completing this journal-based SA-CME activity, participants will be able to:*

- Describe the advantages and risks of NSM and what is involved surgically.
- Discuss the pertinent imaging findings at mammography, US, and breast MRI that can preclude candidacy for NSM in patients with newly diagnosed malignancy.
- Recognize the normal MRI appearance of flap reconstruction in patients who have undergone NSM and of common benign complications and recurrent disease.

*See rsna.org/learning-center-rg.*

## Introduction

Although breast-conserving surgery remains the most frequently performed surgical treatment of breast cancer, the mastectomy rate is increasing (1–3). Nipple-sparing mastectomy (NSM), in particular, has become increasingly popular. The primary concern with NSM is local tumor recurrence. Approximately 10% of breast cancers arise from lactiferous ducts within 2 cm of the nipple, and occult neoplastic nipple involvement occurs in 6%–58% of cases (4,5). Even patients with breast tumors far from the nipple can have associated ductal carcinoma in situ (DCIS) that extends to the nipple.

NSM has been shown to be oncologically safe for appropriate candidates with breast cancer and patients at high risk who are undergoing prophylactic mastectomy (6,7). Reported rates of tumor recurrence in the nipple areolar complex (NAC) are low, at 0%–4% in most studies (3,5,6,8). Authors of nonrandomized studies (9,10) have documented comparable local recurrence outcomes between NSM and skin-sparing or conventional mastectomy. NSM has been associated with improved cosmetic outcomes, with patients undergoing mastectomy reporting better sexual well-being and quality of life and higher satisfaction with postsurgical appearance (11–15). In one study (16), 75% of women experienced the return of some nipple sensation and were satisfied with the pigmentation, projection, and symmetry of their breasts.

## TEACHING POINTS

- Care must be taken at breast MRI not to confuse an inverted nipple with a retroareolar mass, because this may lead to an inappropriate recommendation for biopsy or may preclude a patient from undergoing NSM.
- Labeling of cancers in the retroareolar location at US, mammography, and MRI can result in a misleading TND, with the finding potentially located anywhere from anterior to posterior depth in the breast.
- In a recent study, Wu et al showed that a preoperative TND of less than 1 cm should not preclude NSM as long as no involvement of the NAC is found clinically or at imaging and if retroareolar margins are intraoperatively confirmed to be negative for tumor cells.
- MRI is the most sensitive modality for detection of malignancy involving the NAC and should be performed when NSM is under consideration, even in patients with posterior cancers and those undergoing prophylactic mastectomy. This is routine practice at many institutions.
- Unless it is definitively benign, a new enhancing lesion at breast MRI in patients who have undergone mastectomy and breast reconstruction for cancer is concerning for recurrence, and tissue diagnosis should be obtained.

Prior oncologic guidelines for safe NSM excluded patients with tumors larger than 3 cm, a tumor-to-nipple distance (TND) of less than 2 cm, lymph nodes clinically positive for disease, involvement of the skin, inflammatory breast cancer, Paget disease, prior or planned postmastectomy radiation therapy, and preoperative breast MRI findings suggestive of involvement of the nipple or subareolar tissue (11,17).

Per current National Comprehensive Cancer Network (NCCN) guidelines, NSM is an acceptable surgical option for appropriately selected patients who include those with early-stage or locally advanced breast cancers, those with small-to-moderate breast volume and an acceptable preoperative nipple position of minimal-to-moderate ptosis, imaging findings indicating no nipple or subareolar involvement, nipple margins assessed at pathologic examination and found to be negative for cancer, and the absence of nipple discharge and Paget disease (18,19). In a recent study, Wu et al (9) found that NSM should not be contraindicated, even when TND is less than 1 cm at imaging, provided that no involvement of the NAC is found clinically and retroareolar margins are confirmed to be negative for tumor cells. Controversy over the ideal TND persists because decreased TND has been associated with an increased local-regional recurrence rate (25.0% in patients with a TND < 1 cm vs 2.4% in those with a TND > 1 cm after an average follow-up time of 48 months) in other studies (20).

NSM is considered safe in patients with genetic mutations, including those of the *BRCA* gene, who are undergoing prophylactic mastectomy (21). Nipple preservation is also being performed after neoadjuvant therapy regardless of tumor stage and nodal status, the presence of multifocal versus multicentric disease, tumor histologic results, or the presence of in situ disease and has been shown to be oncologically safe in appropriately selected patients (Fig 1) (11,22,23). Although the complication rates are higher and nipple retention rates are lower, prior radiation or planned postmastectomy radiation therapy is also not a contraindication to NSM (7,11,17–20) (Table 1). Currently, the only absolute contraindications for NSM are clinical or imaging findings of malignancy involving the nipple and inflammatory breast cancer (11).

Imaging findings that should raise suspicion for malignancy involving the NAC include new nipple retraction at mammography and MRI, calcifications suspicious for disease close to or in the nipple at mammography, and loss of the normal zonal anatomy of the nipple at MRI. Additional factors associated with occult nipple involvement include nipple discharge, larger and higher-grade tumors, hormone receptor–negative or human epidermal growth factor receptor 2–positive (HER2+) subtypes, an extensive intraductal component, multifocality, multicentricity, axillary lymph node status, and lymphovascular invasion (5,24,25).

As guidelines evolve, imaging continues to be vital to appropriate patient selection for NSM. Radiologists should have a thorough knowledge of the anatomy of the NAC and its normal and abnormal appearances at mammography, US, and MRI. Radiologists must be meticulous in their assessment and reporting of findings that may affect candidacy for NSM.

## Brief Surgical Overview of NSM

Skin-sparing mastectomy and NSM are part of a category referred to as conservative mastectomy, which continues the trend toward minimum effective treatment and oncologic plastic surgery that combines tumor removal and preparation of a skin flap for immediate reconstruction (26). The goal is to provide local tumor control and cosmesis (20).

Traditional mastectomy involves removal of the skin overlying the breast and the NAC. Skin-sparing mastectomy preserves the skin of the breast but removes the NAC. NSM is similar to skin-sparing mastectomy but preserves the entire NAC and breast skin envelope, with removal of the mammary tissue (5). The pocket created between the skin and the pectoralis muscle is filled with an implant, tissue expander, or native tissue (19).

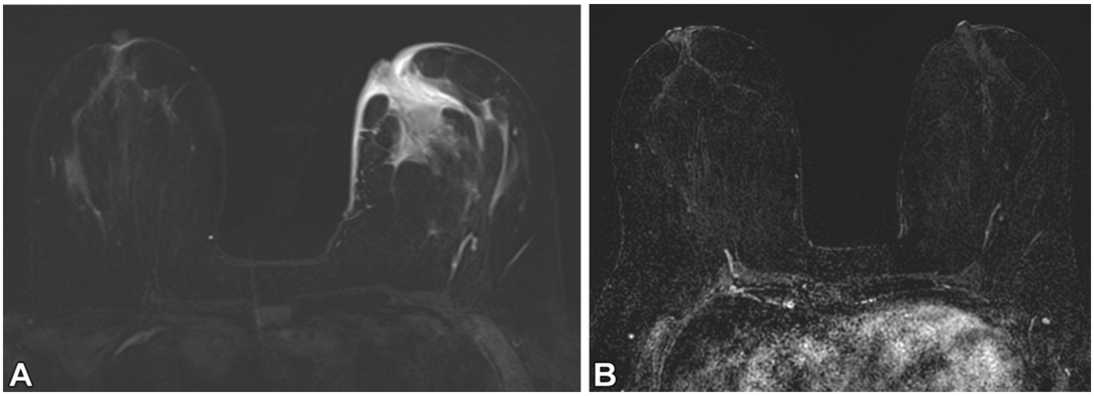


**Figure 1.** Biopsy-proven estrogen receptor (ER)–positive, progesterone receptor (PR)–negative, human epidermal growth factor receptor 2 (HER2)–positive (ER+, PR–, HER2+) invasive ductal carcinoma (IDC) of the left breast in a 42-year-old woman. **(A)** Axial MR image shows a 4.7-cm enhancing mass in the left breast in the 11-o'clock position, 4 cm from the nipple, corresponding to the known cancer with nonmass enhancement extending into the nipple and associated thickening of the skin of the nipple areolar complex (NAC). The patient underwent neoadjuvant therapy. **(B)** Axial MR image obtained after neoadjuvant therapy shows absence of the mass seen in **A**, NAC enhancement, and skin thickening. The patient underwent left lumpectomy with preservation of the nipple, and surgical margins were negative for IDC and DCIS.

**Table 1: Clinical-Pathologic Factors for Appropriateness of NSM: New and Old Guidelines**

| Factor | New Guidelines | Old Guidelines |
|---|---|---|
| TND (cm) | >1 | >2 |
| Node status | Can be performed with positive nodes | Clinically negative nodes |
| Tumor size | Considered in patients with large tumors, good response to neoadjuvant therapy, and negative intraoperative retroareolar tissue sampling results | Tumor size <3 cm, no skin involvement |
| Breast size | Large or ptotic breasts not excluded | Large or ptotic breasts often excluded |
| Contraindications | No inflammatory breast cancer, Paget disease, nipple involvement in tumor, or nipple discharge | No inflammatory breast cancer, Paget disease or nipple involvement in tumor, or nipple discharge |
| Intraoperative section | Negative intraoperative frozen or permanent section from the nipple base | Negative intraoperative frozen or permanent section from the nipple base |
| Safety | Safe in patients with genetic mutations (including *BRCA*) and for prophylactic mastectomy | Unknown safety in patients with genetic mutations |
| Preoperative imaging | Breast MRI negative for malignancy involving the NAC strongly suggested | Breast MRI negative for malignancy involving the NAC strongly suggested |

Note.—Appropriateness criteria for NSM continue to evolve and some indications remain controversial, but the literature currently support these guidelines.

The surgeon aims to remove all glandular tissue while preserving enough subcutaneous fat to prevent compromise of the blood supply, which increases the risk of flap necrosis. The ideal NSM flap includes the entire skin and subcutaneous fat superficial to the breast fascia (20). The surgeon also performs a radical dissection of the retroareolar tissue (19). Surgeons can obtain intraoperative frozen specimens or shave biopsies from the base of the nipple to assess its margins. Some surgeons have abandoned this step and instead send nipple margins for final histologic examination, because both types of biopsy are prone to inaccuracies (11,27). Patients who undergo NSM are counseled preoperatively that if the nipple margin is positive for disease, the standard of care is removal of the NAC (Fig 2).

The complications of NSM are similar to those of other postmastectomy reconstructions and include infection, wound dehiscence, seroma, hematoma, fat necrosis, and flap necrosis (17). Another potential complication of NSM is NAC necrosis. In one study (28), the rate of nipple necrosis with NSM was estimated to be approximately 11%.

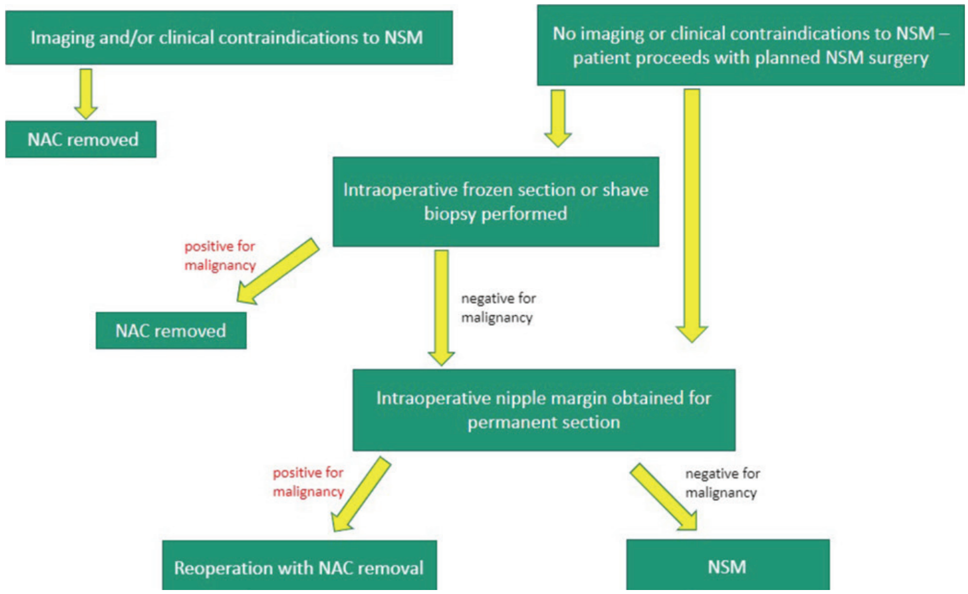



**Figure 2.** Flowchart shows the surgical planning algorithm for NSM.

## Imaging the NAC

The normal NAC is composed of sensory nerve endings, smooth muscle, lymphatic tissue, and milk-secreting sebaceous glands (ie, Montgomery glands) that drain into Morgagni tubercles, (ie, the small papules on the areola) (29). Normal nipples have a variety of shapes and sizes, including everted (75%), flat (23%), and inverted (2%) nipples, and they show several distinct enhancement patterns at MRI (30,31). Accurate imaging and identification of findings in the NAC and retroareolar area, which extend to the base of the nipple, pose challenges.

### Mammography

The role of mammography in evaluating the NAC and detecting malignancy remains crucial, despite the mammographic limitations of breast density, overexposure of tissue, and occasionally, oblique positioning of the nipple, which can obscure a mass (31). When the breast is poorly positioned at mammography, retroareolar masses can be missed or obscured by the nipple. Masses may also be incorrectly localized closer to the nipple than they are (Fig 3).

Calcifications should be evaluated with magnification views and managed with the same Breast Imaging Reporting and Data System (BI-RADS) algorithms for calcifications elsewhere in the breast (32). Comparison with prior imaging can show changes such as new nipple inversion or retraction, which raises the concern for malignancy (30). For optimal mammographic evaluation, the breast must be positioned correctly, with the nipple imaged in profile on at least one view. When the nipple cannot be imaged in profile with routine views, additional projections may be obtained, including one centered on the anterior one-third of the breast. For patients with inverted nipples, the nipples should be tangential and symmetric (33).

### Ultrasonography

US is an effective supplemental tool to assess the NAC that does not expose the patient to ionizing radiation and provides good spatial resolution of superficial tissue (33). US may be targeted toward any palpable or visualized abnormality to provide real-time information. US of the NAC provides higher accuracy than does mammography but is limited by user technique and is complicated by features of the nipple that can make visualization difficult (31,34). At US, the nipple often demonstrates posterior acoustic shadowing that reflects the fibrous composition of the nipple and the confluence of structures including lactiferous ducts, with interfaces that can obscure abnormalities (29). The nipples are usually homogeneously echogenic, with variation attributed to differing amounts of muscle and connective tissue. Radiating linear hypoechoic or isoechoic structures represent ductules and may appear as echogenic dots (29).

Appropriate technique and awareness of ductal anatomy are essential for obtaining diagnostic images. One technique is to use gel or a standoff pad to remove artifacts caused by air bubbles trapped around the nipple (29). Other techniques include “two-handed compression” and the “rolled-nipple” maneuver (Fig E1). With the two-handed compression technique, the operator compresses the NAC with the free hand to distinguish whether an abnormality is an intraductal mass or debris. A mass does not resolve, whereas secretory debris dissipates. The rolled-nipple technique involves placing the transducer parallel to the duct of

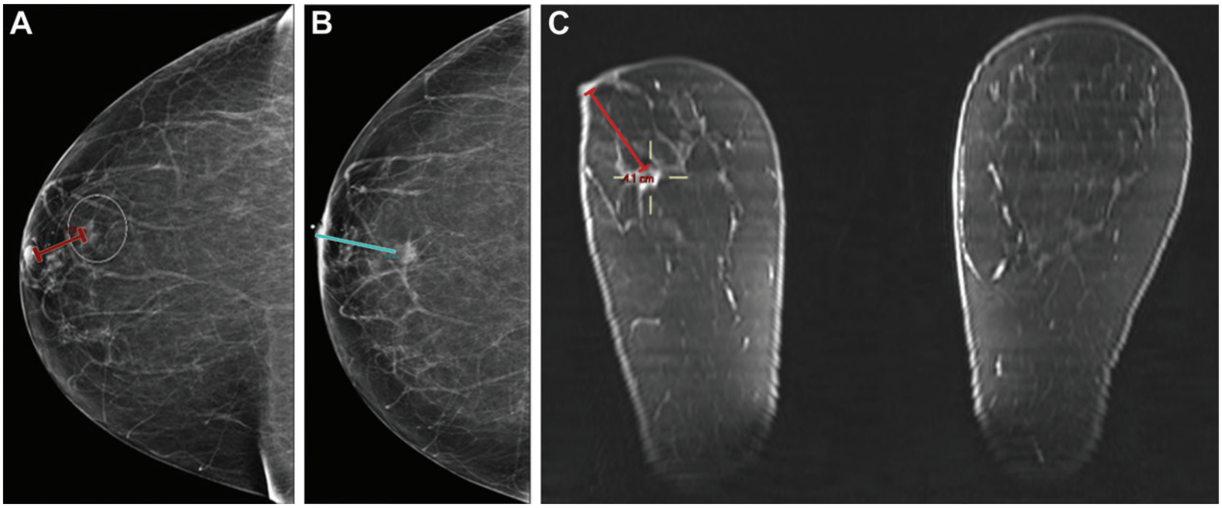



**Figure 3.** The importance of accurate measurement of tumor-to-nipple distance (TND) in a 59-year-old woman with a breast mass. **(A)** Initial craniocaudal mammogram with suboptimal oblique positioning of the nipple shows a 1.0-cm subtle mass (circle) 2.0 cm posterior to the nipple, with the TND indicated (red line). **(B)** Subsequent craniocaudal mammogram with compression over the anterior breast and the nipple in profile shows the mass to be 3.4 cm posterior to the nipple, with the TND indicated (blue line). **(C)** Multiplanar reconstructed MR image obtained after US-guided biopsy shows that the mass is actually 4.0 cm posterior to the nipple, with the TND indicated (red line).

interest. The operator uses the free hand to apply pressure and "roll" the NAC to visualize the area of interest (33,34).

Accurate identification of findings in the NAC region at US also poses a challenge because of interobserver variation. For example, the same finding in the retroareolar location may appear different, depending on how the probe or breast is positioned. Differences in equipment can present challenges for consistent evaluation. Some institutions are using newer technology such as elastography, contrast-enhanced US, three-dimensional US, automatic breast US, and computer-aided detection to overcome some of the challenges of traditional breast US (35).

## Breast MRI

MRI is the most sensitive imaging modality for evaluation of the NAC because of its superior soft-tissue contrast and the addition of contrast enhancement (31). At breast MRI, nipple enhancement is best assessed during the first subtraction series. Since subtraction images are prone to misregistration artifacts related to patient movement between precontrast and contrast-enhanced imaging, the pre- and postcontrast fat-saturated sequences should also be carefully evaluated. Confirmation or exclusion of a finding on images from these sequences can offer the radiologist increased confidence in the diagnosis (31).

Most nipples have a characteristic three-layered appearance and display *(a)* a thin rim of intense superficial linear enhancement at the level of the skin that should be smooth and not thicker than 2 mm; *(b)* a nonenhancing zone below the superficial linear enhancement, which is the dermis, containing connective tissue and sebaceous glands; and *(c)* internal nipple enhancement extending below the nonenhancing zone to the nipple base, which can be linear (20%) or patchy (68%) (31). Internal nipple enhancement is typically equal to or lower than the background parenchymal enhancement. Nipple enhancement is categorized as minimal, mild, moderate, or marked and is compared with the background parenchymal enhancement in the ipsilateral breast (31) (Fig 4).

Although it is more commonly symmetric, normal nipple enhancement can be asymmetric. Areolar thickening is occasionally seen and is considered normal when it is smooth, bilateral, and symmetric (31). The nipples can be everted, flat, or inverted, and this appearance can be symmetric or asymmetric. Both inversion and retraction can be congenital or acquired and unilateral or bilateral. A benign inverted nipple, one in which the entire nipple is pulled in, must be differentiated from an abnormally retracted nipple, in which a portion of the nipple is pulled in by an underlying mass. Care must be taken at breast MRI not to confuse an inverted nipple with a retroareolar mass, because this may lead to an inappropriate recommendation for biopsy or may preclude a patient from undergoing NSM (Fig 5).

## Labeling and Correlating Multimodality Findings

Positioning of the patient varies from one imaging modality to another. Breast MRI is performed with the patient in a prone position, US is performed with the patient in a supine position, and

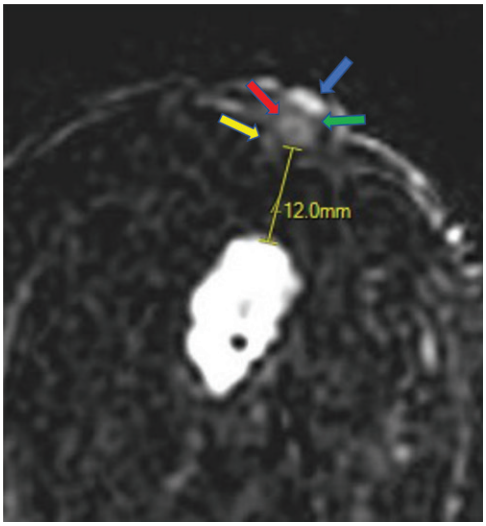



**Figure 4.** Normal nipple enhancement in a 47-year-old woman with an irregular heterogeneously enhancing mass in the right retroareolar location, nonenhancing parenchyma, and a TND of 12 mm. Axial subtraction MR image shows superficial linear enhancement (blue arrow), the nonenhancing zone (green arrow), internal nipple enhancement (red arrow), and the base of the nipple (yellow arrow).

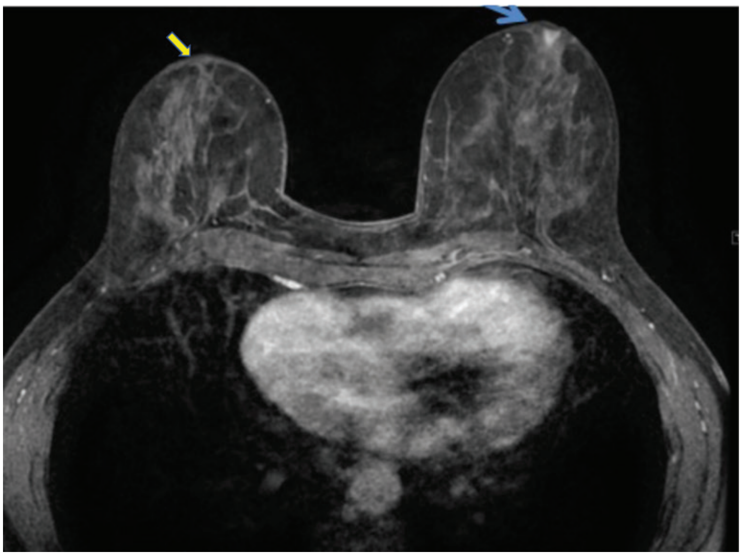


**Figure 5.** Variations in nipple size, shape, and enhancement in a 55-year-old woman. Axial contrast-enhanced T1-weighted MR image shows a flat right nipple (yellow arrow), an inverted left nipple (blue arrow), and asymmetric nipple enhancement.

mammography is performed with the patient in an upright position (Fig 6) (36). As a result, the TND may be discordant among mammography, MRI, and US (36). A finding labeled in the retroareolar location at US (Fig 7A, Fig E2A) may be at the middle or posterior depth at mammography and MRI (Fig 7B, Fig E2B) and have a TND acceptable for NSM. Labeling of cancers in the retroareolar location at US, mammography, and MRI can result in a misleading TND, with the finding potentially located anywhere from anterior to posterior depth in the breast. To communicate the location of a finding most accurately, the American College of Radiology (ACR) BI-RADS Atlas (37) recommends describing the quadrant, clock face position, depth, and TND. Descriptions such as "retroareolar" or "central to the nipple" can be used, when applicable. The depth of a finding should be reported as the anterior, middle, or posterior one-third of the breast. Recording the TND provides an additional and more precise indication of the depth of the lesion. Precise labeling helps to reduce discrepancies in reported TND among mammography, US, and MRI. Even with meticulous technique, the question of NAC involvement in malignancy may remain after US and mammography (Figs 8, 9). Breast MRI, with its volumetric imaging, has been found to be beneficial before NSM for better visualization of the retroareolar tissue, identification of malignancy, and accurate measurement of the TND (38,39). Sagittal or oblique reformatted or multiplanar reconstruction images can be helpful to visualize the mass in the same plane as that of the nipple for accurate measurement of the TND.

## Candidacy for NSM

Choosing the best candidates for NSM before surgery is essential for decreasing the risk of local-regional recurrence and the need for

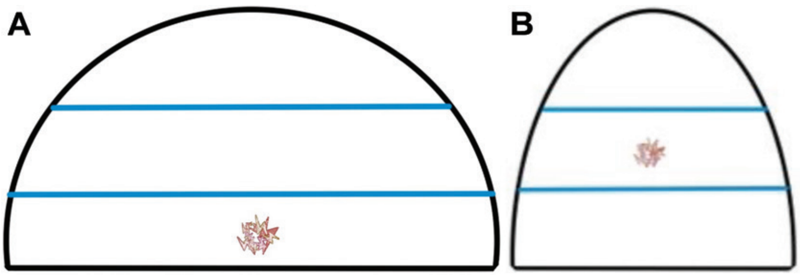



**Figure 6.** Illustrations show how the location of a mass in the breast can vary as a result of the positioning of the breast at US and mammography. **(A)** At US, with the patient in a supine position, gravity results in compression of the breast tissue toward the chest wall, and the mass (red) appears in the posterior one-third of the breast. **(B)** At mammography, the mass (red), with the breast pulled up and out, appears in the middle one-third of the breast.

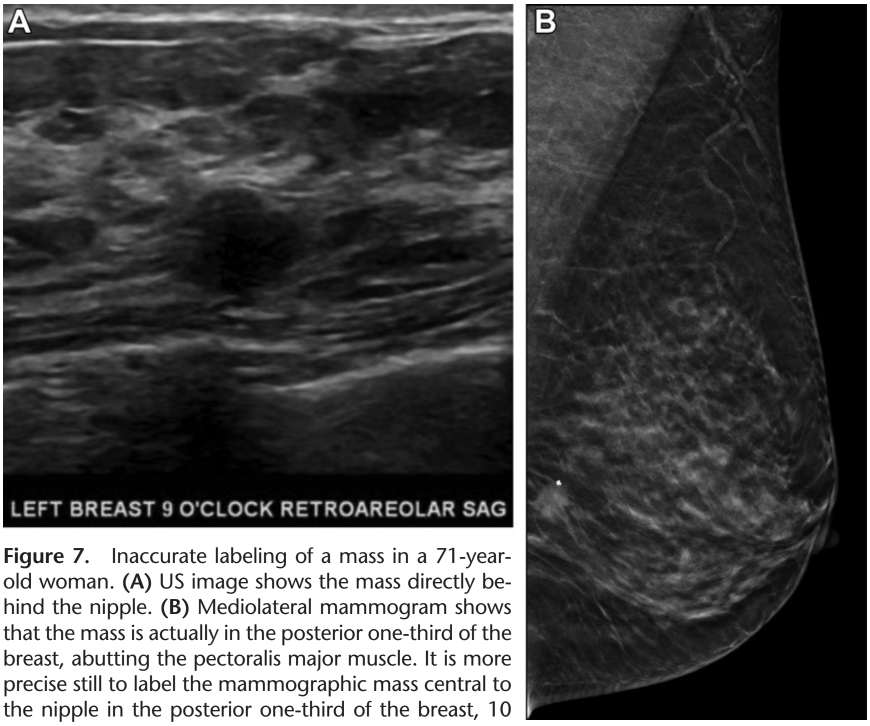



**Figure 7.** Inaccurate labeling of a mass in a 71-year-old woman. **(A)** US image shows the mass directly behind the nipple. **(B)** Mediolateral mammogram shows that the mass is actually in the posterior one-third of the breast, abutting the pectoralis major muscle. It is more precise still to label the mammographic mass central to the nipple in the posterior one-third of the breast, 10 cm deep to the nipple. Labeling the mass as retroareolar did not accurately communicate the depth of the mass in the breast, and as a result, the surgeon questioned whether the patient was a candidate for NSM.

potential additional surgical procedures in those found to have disease in the nipple on the basis of intraoperative sampling. Nipple excision during a second surgical procedure can negatively affect the cosmetic outcome and increase the risk of necrosis (27).

## Mammographic Findings that Preclude NSM

In all patients with new or suspected malignancy, mammograms should be carefully reviewed for findings that could preclude candidacy for NSM. Factors that are highly suggestive of malignancy involving the NAC at mammography include a TND of less than 1 cm, new nipple inversion or retraction, and areolar distortion (29). NAC asymmetry, retroareolar masses, microcalcifications, and skin thickening have all been associated with malignant involvement (40,41) (Fig 10).

TND measurements should be made from the anterior aspect of a mass or calcifications on the mammogram to the base of the nipple. Although masses found in the retroareolar location are most commonly benign and include papillomas, fibroadenomas, adenomas, and complicated cysts, they can be malignant. In patients with a known malignancy in a different part of the breast, tissue diagnosis of retroareolar masses should be established in those considering NSM.

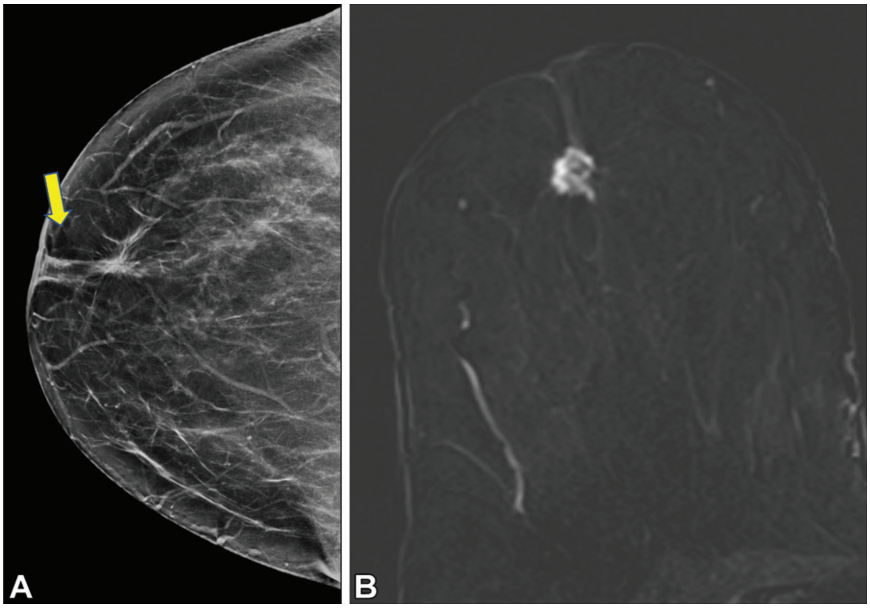



**Figure 8.** Well-differentiated IDC and ductal carcinoma in situ (DCIS) in a 73-year-old woman who underwent US-guided biopsy of a mass in the right retroareolar location. **(A)** Mammogram shows the mass, with possible extension (arrow) to the nipple. **(B)** Axial subtraction MR image shows the index malignancy in the retroareolar location, with 2.2 cm of nonenhancing, uninvolved breast tissue between the mass and nipple, making the patient a candidate for NSM.

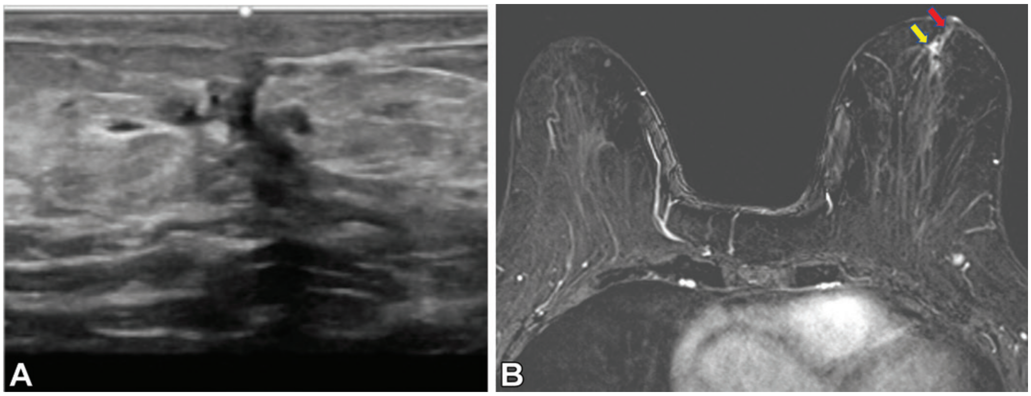



**Figure 9.** Invasive mammary carcinoma with ductal and lobular features in a 75-year-old woman who showed architectural distortion in the left medial retroareolar location on screening digital breast tomosynthesis images (not shown). **(A)** Gray-scale US image shows a 0.9-cm irregular hypoechoic mass corresponding to the mammographic finding, with dilated retroareolar ducts and possible extension to the nipple. US-guided biopsy results (not shown) were positive for invasive mammary carcinoma with ductal and lobular features. **(B)** Axial subtraction MR image shows the enhancing mass (yellow arrow) and linear nonmass enhancement (red arrow) extending to the nipple.

New nipple inversion or retraction may be subtle and should be carefully assessed. New unilateral nipple inversion and distortion of the areola are common signs of malignancy, while long-standing inversion, bilateral slowly progressing nipple inversion, and slitlike nipple inversion are less concerning (29,30) (Fig 11).

NAC calcifications are found in some patients. They are typically benign and include dermal calcifications, those in the Montgomery glands, and those within hair follicles (29). Benign dermal calcifications are seen in the NAC of patients with a history of breast surgery, commonly in those who have undergone reduction mammoplasty. Benign calcifications may also be seen within a mass (eg, coarse "popcorn" calcifications in a degenerating fibroadenoma). Calcified cutaneous keratin growths can occur above the NAC skin surface and are associated with benign and malignant processes including Paget disease and squamous cell carcinoma (29). Calcifications in the NAC may be malignant and include invasive cancer and DCIS. Magnification mammographic views, tomosynthesis, and tangential views can be helpful in determining whether calcifications appear benign or have morphologic characteristics that are suspicious for cancer (Figs 12–14). For patients in whom calcifications remain indeterminate after additional imaging, biopsy may be necessary for accurate determination of candidacy for NSM.

After neoadjuvant chemotherapy, malignant calcifications sometimes decrease in number but may persist at mammography (42). When they are persistent on mammograms, they do not

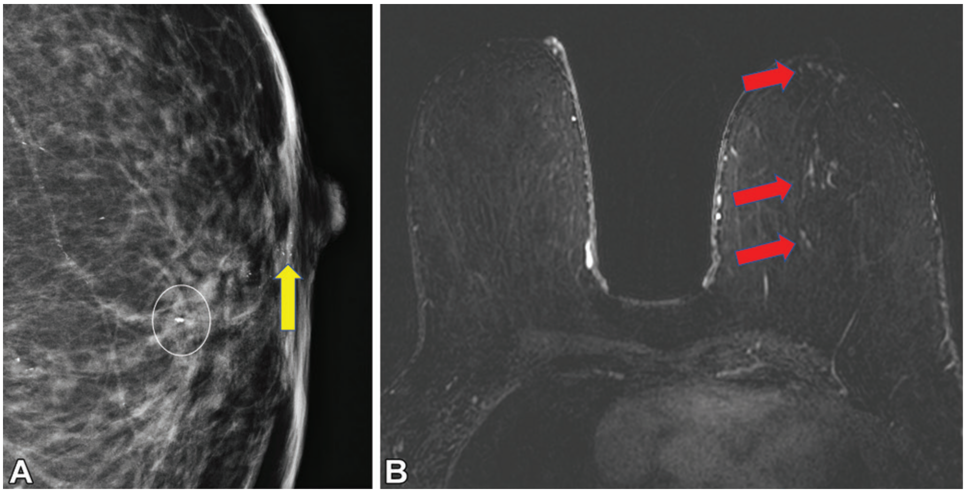



**Figure 10.** Intermediate nuclear-grade DCIS in a 62-year-old woman who underwent stereotactic biopsy of coarse heterogeneous microcalcifications in the left anterior central breast. **(A)** Postprocedural mediolateral mammogram shows that although the biopsy clip was more than 1 cm from the nipple (circle), the calcifications extend to within 0.3 cm of the base of the nipple (yellow arrow). **(B)** Axial pretreatment subtraction breast MR image shows multiple areas of linear nonmass enhancement with interspersed nonenhancing parenchyma that extends 9 cm posteriorly from the base of the nipple (red arrows). This patient was not considered a suitable candidate for NSM.

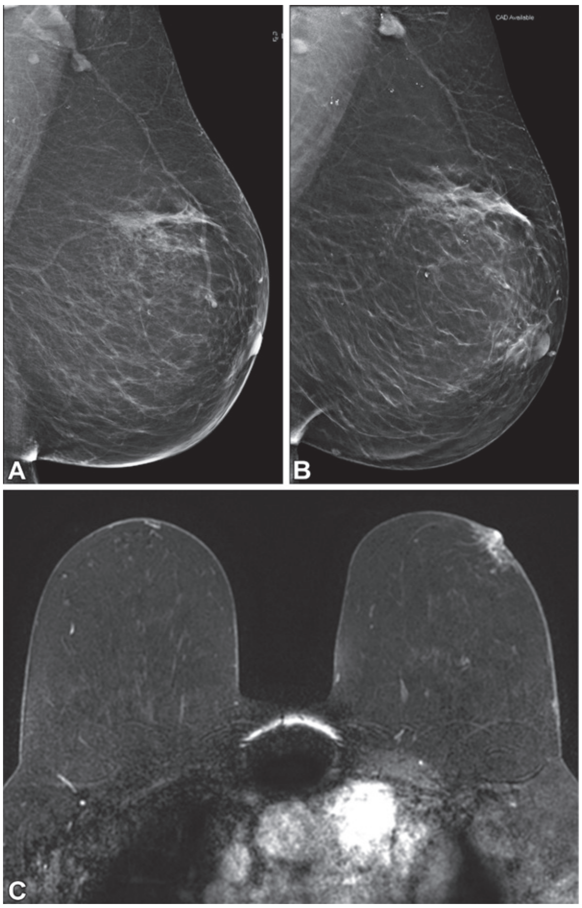



**Figure 11.** IDC in a 67-year-old woman who reported new palpable hardening in the left retroareolar location. **(A)** Mediolateral oblique mammogram shows a normal appearance of the left nipple and retroareolar region. **(B)** Mediolateral two-dimensional synthesized mammogram of the left breast obtained 4 years later shows subtle inversion of the left nipple and increased attenuation in the retroareolar location when compared with the appearance in **A**. The patient underwent US-guided core biopsy of a mass in the left breast in the 3-o'clock position, 4 cm from the nipple (not shown), which showed IDC. **(C)** Axial fat-saturated contrast-enhanced MR image shows abnormal enhancement of the NAC, nipple retraction, and areolar distortion. The biopsy-proven mass in the left breast was also visualized (not shown).

necessarily indicate residual disease, but excision of all indeterminate or suspected calcifications remains standard practice (43).

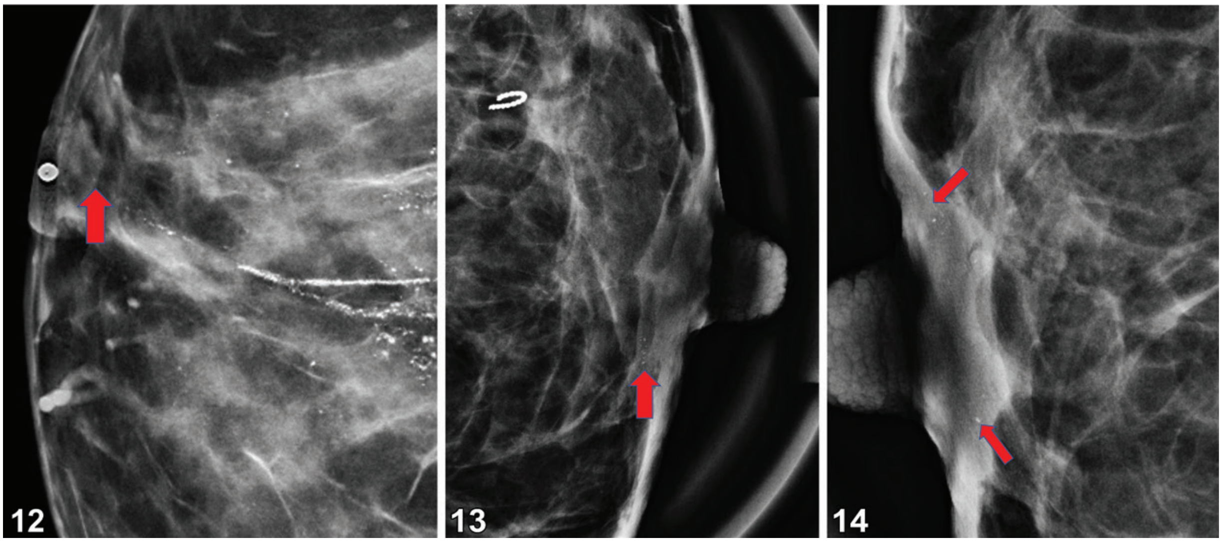
## US Findings that Preclude NSM

Malignancy in the retroareolar location at US can be nonspecific and can mimic infection. Although discrete intraductal or retroareolar masses may be seen, malignancy may also appear as heterogeneous or hypoechoic areas, skin thickening, or dilated lactiferous ducts with debris or intraductal masses (29).

Even with meticulous mammographic and US technique, limitations in the evaluation of the NAC and retroareolar location for malignancy persist; therefore, most surgeons who are considering NSM for a patient request a breast MRI examination (Fig 15).

## MRI Findings that Preclude NSM

Although the overall utility of routine preoperative breast MRI is controversial, contrast-enhanced breast MRI is the most useful modality for accurately predicting NAC involvement in malignancy (44,45). It can guide appropriate patient selection for NSM and is required by many surgeons. Contrast-enhanced breast MRI has higher sensitivity than do mammography and US for diagnosis and assessment of cancers involving the NAC and the retroareolar location (46). MRI can help to differentiate the normal nipple and tumor confined to the retroareolar tissue from a tumor involving the NAC (47).




**Figures 12–14.** Spot magnification mammograms in two patients. **(12)** Craniocaudal spot magnification mammogram in a 41-year-old woman shows segmental pleomorphic calcifications, including casting calcifications that extend to within 1 cm of the nipple (arrow) in the posterior to the anterior one-third of the breast. Biopsy results showed DCIS. **(13)** Mediolateral spot magnification mammogram in a 39-year-old woman with biopsy-proven IDC and DCIS in the anterior-superior lateral left breast shows additional grouped calcifications (arrow) in the left inferior medial retroareolar location, raising concern for the clinical appropriateness of NSM desired by the patient. **(14)** Mediolateral spot magnification mammogram of the right breast of the same patient as in Figure 13 shows benign calcifications (arrows) in the skin of the areola.

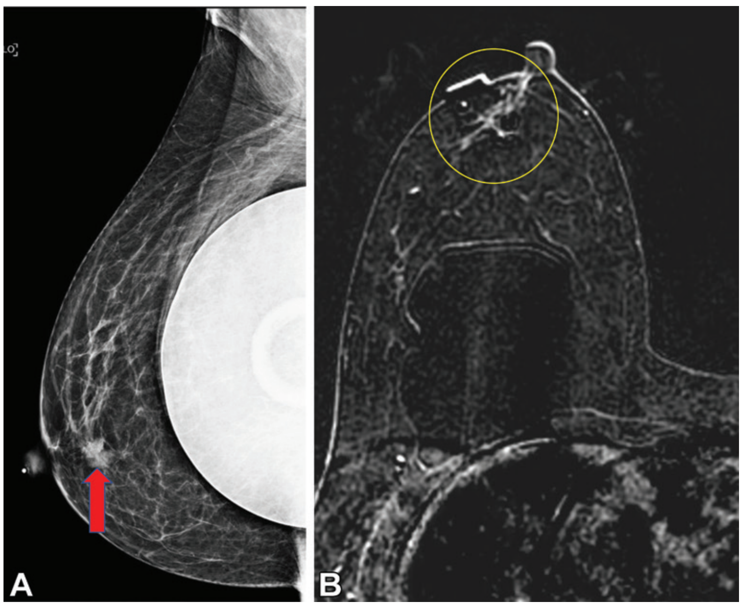



**Figure 15.** Evaluation of candidacy for NSM in a 61-year-old woman with a history of bilateral breast augmentation with saline implants. **(A)** Mediolateral oblique mammogram shows a new mass in the right retroareolar location (arrow). US (not shown) showed an irregular hypoechoic mass, with no visible extension to the nipple or skin. **(B)** Subsequent axial subtraction MR image shows nonmass enhancement from the mass that is contiguous to the base of nipple (circle). Therefore, this patient was not considered a candidate for NSM. Although both modalities allowed visualization of the mass at 1 cm posterior to the base of the nipple, MRI is superior for assessment of the retroareolar location, especially in patients who are considering NSM.

Authors of several studies have investigated the use of breast MRI for prediction of neoplastic involvement of the NAC (25,44,45) (Table 2). In a study of 81 patients with breast cancer, Sakamoto et al (44) reported that unilateral enhancement of the nipple in continuity with the index lesion at MRI was a statistically significant predictor of malignant involvement of the NAC. In addition, they found that skin enhancement, enhancement of the ductal area, a pattern of rim enhancement, and diffuse enhancement of the breast may suggest a higher risk of neoplastic involvement of the nipple (44).

Moon et al (45) studied the accuracy of breast MRI to predict malignant involvement of the nipple in 51 patients with breast cancer. At preoperative breast MRI, they found a statistically significant correlation between neoplastic involvement of the nipple and NAC enhancement, nipple inversion or retraction, and periareolar skin thickening (45). They concluded that breast MRI showed sensitivity of 93.8% and specificity of 85.7% for detection of malignancy involving the NAC (45).

Piato et al (25) reported that enhancement of the index lesion compared with that of the

**Table 2: MRI Findings of Involvement and Uninvolvement of the NAC in Malignancy**

| | |
|---|---|
| Involved in malignancy | |
| | Absence of enhancement between the cancer and nipple |
| | Distance >1 cm between the cancer and the nipple |
| | Absence of nipple retraction |
| Uninvolved in malignancy | |
| | Presence of enhancement between the cancer and nipple |
| | Abnormal enhancement of the nipple or NAC |
| | Periareolar skin thickening |
| | Nipple retraction, areolar distortion |
| | Diffuse enhancement of the breast |
| | Skin enhancement |
| | Enhancement of the central ductal area |
| | Rim-enhancing appearance of cancer in the anterior central breast |

Note.—Many surgeons consider presurgical breast MRI a requirement in patients with newly diagnosed malignancy who are contemplating NSM. Specific MRI findings are associated with the presence or absence of malignancy involving the NAC, and the interpreting radiologists should carefully assess for them.

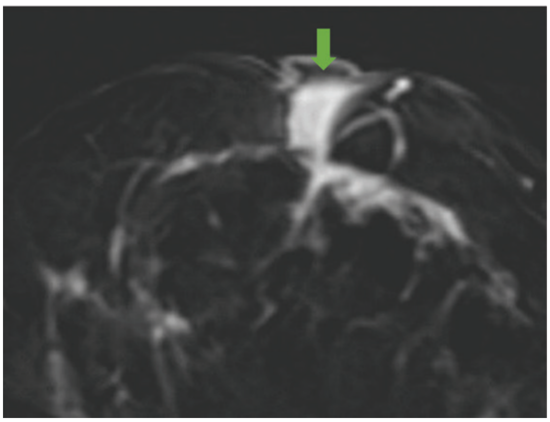


**Figure 16.** Newly diagnosed invasive lobular carcinoma in the right breast in a 45-year-old woman who presented for pretreatment MRI. Axial subtraction MR image shows an enhancing mass (arrow) in the right retroareolar location that extends into the nipple, with associated nipple retraction and areolar distortion. Note the disruption of normal internal nipple enhancement.

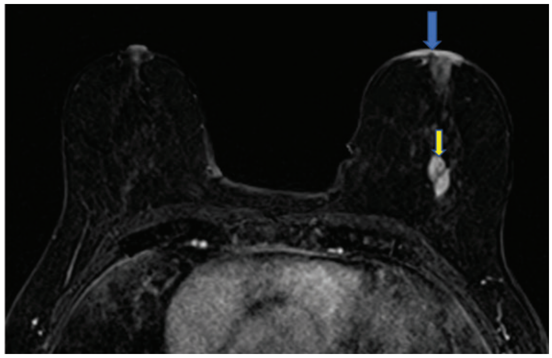


**Figure 17.** Paget disease, DCIS, and IDC in a 78-year-old woman. Contrast-enhanced T1-weighted fat-saturated MR image shows an abnormally enhancing NAC (blue arrow). All normal zones of nipple enhancement are altered. An additional 3-cm enhancing mass (yellow arrow) that is located posterior to the nipple was found at subsequent needle biopsy to be IDC and DCIS.

nipple (odds ratio, 9.25; 95% CI: 3.67, 23.32; $P < .01$) and nipple retraction (odds ratio, 4.57; 95% CI: 1.86, 11.06; $P = .01$) are MRI findings suggestive of neoplastic involvement of the NAC (25). They also showed that the absence of enhancement between the index lesion and the nipple and the associated absence of nipple retraction were the two most important imaging findings at breast MRI that exclude malignancy involving the NAC. The negative predictive value of the combination of absence of these two characteristics was 83.3% (95% CI: 76.5%, 88.8%) (25). Authors of other studies (47,48) have shown that breast MRI, TND, enhancement of the index malignancy compared with that of the nipple, and nipple retraction have been found to be the factors that are most significantly associated with malignancy involving the NAC (Fig 16) .

In a recent study, Wu et al (9) show that a preoperative TND of less than 1 cm should not preclude NSM as long as no involvement of the NAC is found clinically or at imaging and if retroareolar margins are intraoperatively confirmed to be negative for tumor cells. In this study, the authors acknowledged that a TND of less than 1 cm remains controversial, and that even if the results of intraoperative sampling of the retroareolar margins are negative, missed lesions may remain in the papilla of the nipple, which can be a nidus for recurrence (9).

At MRI, abnormal nipple enhancement is seen as a breach of normal nipple zone enhancement, disruption of normal nipple structures, or nodular or irregular enhancement along the posterior borders of the nipple (Fig 17). Loss of normal internal nipple enhancement is suspicious for malignancy. With tumors in the breast, invasion progresses anteriorly from the base of the nipple and can be subtle, leaving most of the normal zones of enhancement intact (31).

TND is well evaluated at MRI. A TND of less than 1 cm indicates a high possibility of malignancy in the NAC. TND should be clearly noted in the report, along with the presence or absence of nipple retraction and tumor invasion into the nipple and NAC. These findings are crucial to

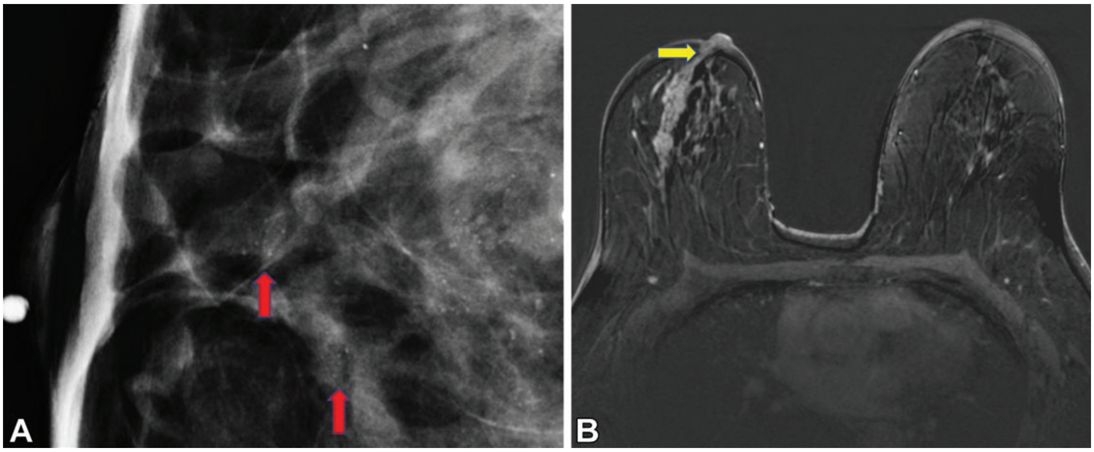



**Figure 18.** Biopsy-proven DCIS in a 53-year-old woman. **(A)** Mammogram shows calcifications that extend to within 1.2 cm of the nipple (arrows). Under current criteria, disease greater than 1 cm from the nipple at mammography does not preclude the patient from undergoing NSM. **(B)** Axial pretreatment T1-weighted MR image shows asymmetric thick linear nonmass enhancement that is extending into the right nipple (yellow arrow). On the basis of these MRI findings, the patient is not a candidate for NSM.

assess whether a patient is a good candidate for NSM (Fig 18). MRI is the most sensitive modality for detection of malignancy involving the NAC and should be performed when NSM is under consideration, even in patients with posterior cancers and those undergoing prophylactic mastectomy. This is routine practice at many institutions (Fig 19).

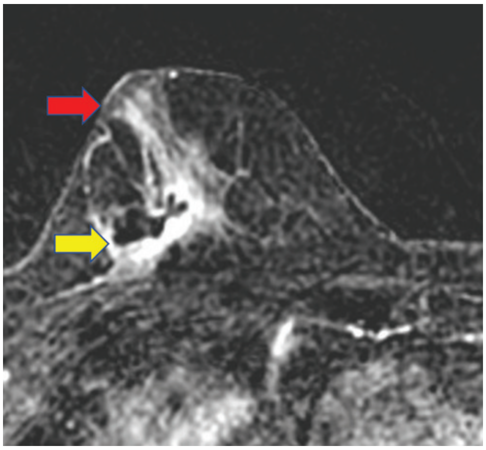


**Figure 19.** Poorly differentiated IDC in an 87-year-old woman who underwent US-guided biopsy of a mass at the 9-o'clock position in the right breast. Axial subtraction MR image shows the biopsy-proven cancer and a small hematoma in the outer posterior right breast (yellow arrow). Although the mass was posterior, segmental nonmass enhancement extends to the nipple, with extension of the malignancy into the nipple papilla (red arrow) and disruption of the normal internal nipple enhancement zone.

## Normal Appearance after NSM and Reconstruction versus Recurrence at Breast MRI

Current NCCN guidelines do not include routine imaging of reconstructed breasts with mammography or breast MRI (18). The ACR Appropriateness Criteria advise that screening mammography may be appropriate in patients with breast cancer who have undergone autologous reconstruction (with or without an implant). The ACR does not recommend screening for patients who have undergone implant reconstruction alone, or for patients who have undergone prophylactic mastectomy and reconstruction (49). However, physical examination can be challenging in patients who have undergone mastectomy and reconstruction, and as NSM becomes more prevalent, more imaging of the reconstructed breast is likely to be performed. Therefore, it is crucial that radiologists identify normal and abnormal findings associated with different reconstructive techniques, including complications, and be able to distinguish them from recurrence (50).

### Flap Reconstruction

Common reconstruction techniques include the use of autologous tissue flaps or implants or a combination of both (51). The imaging appearance of flaps reflects their tissue composition. Flaps can be imaged at mammography, US, and breast MRI, although routine imaging is not performed. Of these modalities, MRI is superior for confirmation of the normal appearance of the flap and differentiation of benign from malignant changes (50). Flaps are composed mainly of fatty tissue and primarily demonstrate the characteristics of fat tissue at mammography, US, and MRI. With myocutaneous flaps, the muscular component of the flap can be visualized at mammography as high attenuation and at MRI as soft-tissue signal intensity, with associated vessels in the posterior region of the

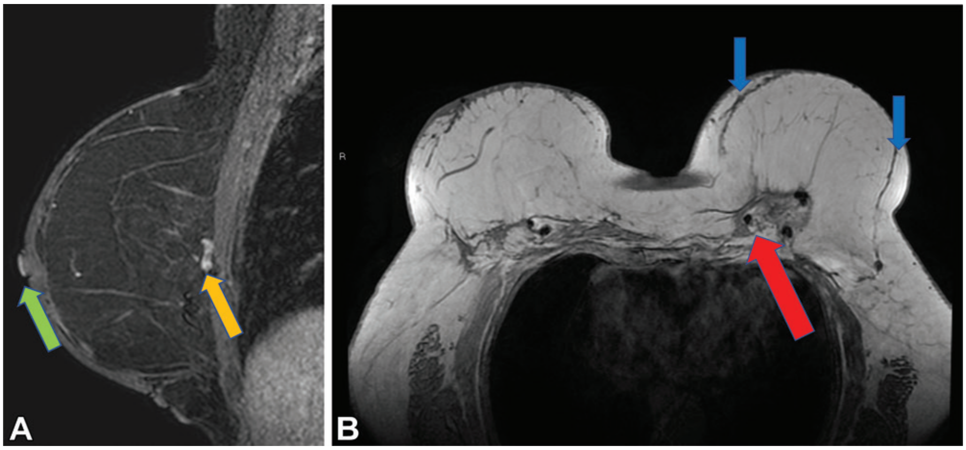



**Figure 20.** Flap reconstruction in a 61-year-old woman. **(A)** Sagittal reformatted MR image shows replacement of normal breast tissue with fat tissue from the lower abdomen, buttocks, or thighs. Free flaps such as deep inferior epigastric perforator flaps show a thin vascular pedicle anastomosed with microsurgery to the internal mammary artery (gold arrow). Note the preserved normal nipple anatomy after NSM. The superficial linear enhancement is well seen (green arrow). **(B)** Axial contrast-enhanced non–fat-saturated T1-weighted MR image shows a thin curved line of intermediate signal intensity that separates native residual breast tissue from the flap (ie, the zone of contact) (blue arrows). Transverse rectus abdominis myocutaneous and latissimus dorsi flaps show atrophied muscle associated with the vascular pedicle along the chest wall (red arrow).

reconstructed breast, anterior to the pectoralis muscle (52). With free flaps, MRI shows a thin vascular pedicle that has been anastomosed with a microsurgical technique to the internal mammary artery (Fig 20A) (53).

Breast MRI is performed to assess for malignancy in the contralateral native breast of some patients at high risk who underwent unilateral mastectomy. It is also sometimes performed to evaluate for recurrence in patients with symptoms after mastectomy and reconstruction. Normal findings vary according to the type of reconstruction performed. The normal MRI appearance of autologous tissue flap reconstructions includes replacement of glandular breast tissue with adipose tissue and the presence of a line of intermediate signal intensity that separates the native tissue from the flap reconstruction (zone of contact) (54). The zone of contact should be thin, without nodularity or focal enhancement (Fig 20B) (54).

The thickness of the native subcutaneous tissue between the skin and the zone of contact varies. Thickness is partly dependent on the patient's body mass index and age. Overweight and older patients have thicker adipose tissue beneath the skin (55). Flap thickness is important to the aesthetic outcome and the complication rate in patients who undergo breast reconstruction, with thicker skin flaps prone to better aesthetics and fewer complications (55). However, increasing skin flap thickness is also associated with increased recurrence; a skin flap thickness of more than 5 mm is associated with a greater number of terminal ductal units and the presence of residual disease (56). Residual glandular tissue in a reconstructed breast after mastectomy may manifest at MRI as dynamic contrast enhancement subjacent to the flap. If residual glandular tissue is seen, this should be stated in the report, and long-term surveillance may be indicated (56). In patients who have undergone NSM, breast contrast-enhanced MRI demonstrates the native nipple, with the classic enhancement pattern including a thin rim of increased superficial signal intensity (56).

## Complications after NSM and Breast Reconstruction

Complications seen within the first 6 months after breast surgery and autologous reconstruction include seromas, hematomas, and infections. US is useful for characterizing fluid collections as simple or complex (57). Seromas are common after surgery and most decrease in size and resolve over time, but a minority persist indefinitely (Fig 21). It is important to note whether a seroma is growing. Seromas that increase in size or complexity raise concern for recurrence (58). The incidence of infection in flaps is 2%–4% in the early postoperative period, and US may be used to distinguish between mastitis and abscesses (59).

Other benign findings include epidermal inclusion cysts and fat necrosis. Fat necrosis occurs more frequently in the immediate to later postoperative period (60), occurs in 94% of the masses during the 1st year after surgery, is the most common cause of palpable and nonpalpable lesions in the reconstructed breast, and is frequently seen at imaging (50,61,62) (Fig 22).

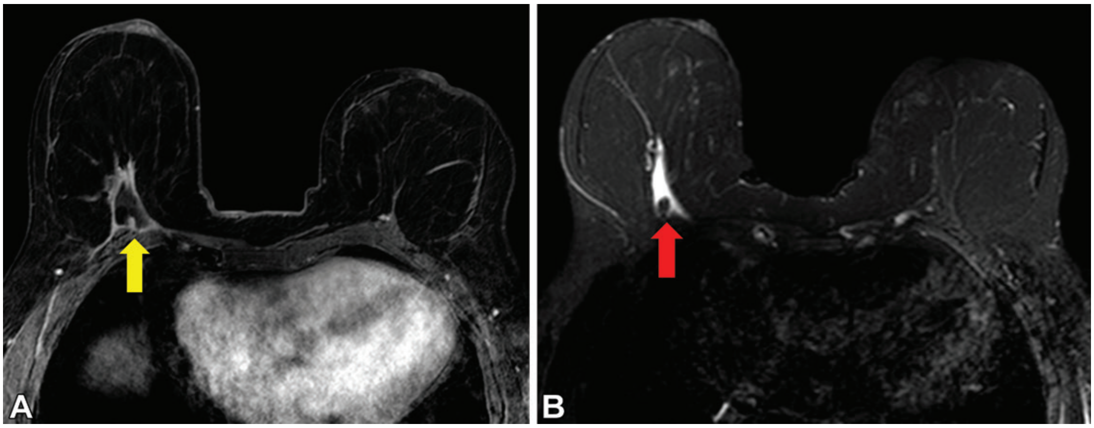



**Figure 21.** Seroma in a 53-year-old woman who underwent bilateral mastectomy and deep inferior epigastric perforator flap reconstruction 7 years previously for cancer of the left breast. **(A)** Axial contrast-enhanced fat-saturated T1-weighted MR image shows a thin rim of peripheral enhancement, surrounding a collection of high-signal-intensity fluid (yellow arrow). **(B)** Axial precontrast fat-saturated T2-weighted MR image shows a well-circumscribed high-signal-intensity fluid collection (red arrow).

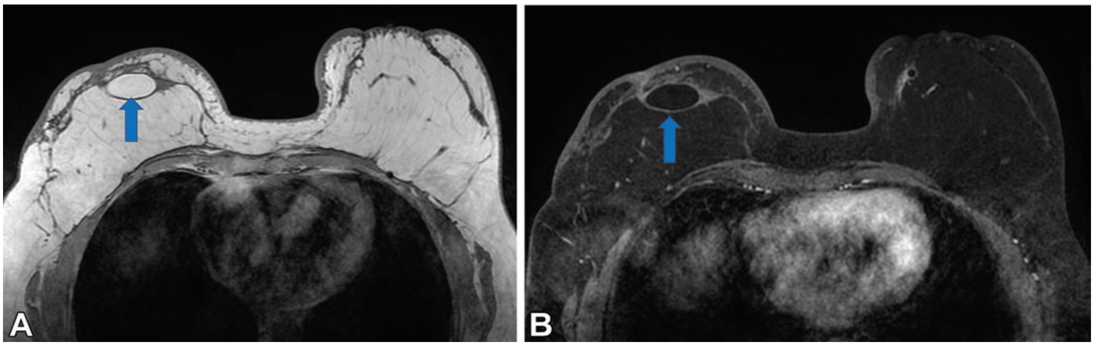



**Figure 22.** Deep inferior epigastric perforator flap reconstruction after bilateral mastectomy for IDC and DCIS of the right breast in a 55-year-old woman. **(A)** Axial precontrast non–fat-saturated T1-weighted MR image shows a high-signal-intensity circumscribed mass in the right upper central reconstructed breast (arrow). **(B)** Axial contrast-enhanced fat-saturated breast MR image shows fat saturation, which is a finding that is consistent with a fat necrosis oil cyst (arrow) along the zone of contact and nonenhancing skin thickening (greater on the right side than that on the left), which is consistent with adjuvant radiation therapy of the right breast.

Areas of fat necrosis are most common along the periphery of autologous flaps, including at the parasternal and superior regions of the chest wall, where the blood supply is less robust (Fig 23) (51,53). Fat necrosis can have a varied appearance and enhancement kinetics at MRI. Areas of signal intensity voids may be seen if calcifications are present. T2-weighted non–fat-saturated images can provide helpful architectural information about MRI findings (53). The diagnosis of fat necrosis at breast MRI can be definitive when internal fat signal intensity is seen at T1-weighted non–fat-saturated MRI. Findings of fat necrosis can also be nonspecific at MRI, and recurrence is included in the differential diagnosis. Correlation of MRI with mammographic findings is often helpful for definitive visualization of fat and confirmation of fat necrosis (54,60). Although it is not performed routinely, mammography of the reconstructed breast sometimes allows confirmation of fat necrosis, obviating the need for biopsy. Some institutions perform annual mammography to evaluate autologous flaps for suspected calcifications and other findings of recurrence. In some cases, the findings may mimic malignancy with all modalities, and tissue sampling may be necessary.

Fat grafting has become an important adjunct to all types of breast reconstruction, especially with prepectoral implant reconstructions, and up to 30% of all breast reconstructions now involve fat grafting. Fat grafting fills in contour irregularities and augments areas to achieve a more natural breast shape (63). Fat grafting is typically not performed at the time of mastectomy, because grafts require robust vascular beds, and surgical manipulation of the flap can cause poor wound healing and necrosis. Patients who are planning to undergo this procedure are told in advance that they will likely require at least a second fat grafting procedure and may benefit from multiple rounds of fat grafting (63).

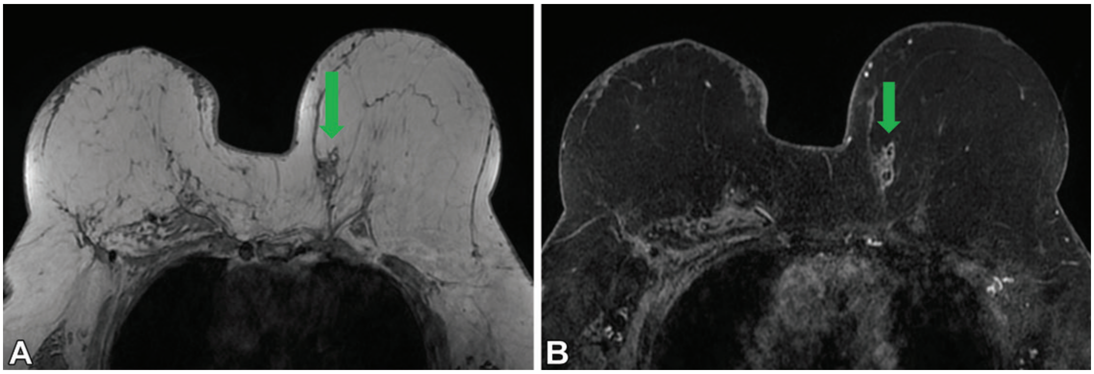
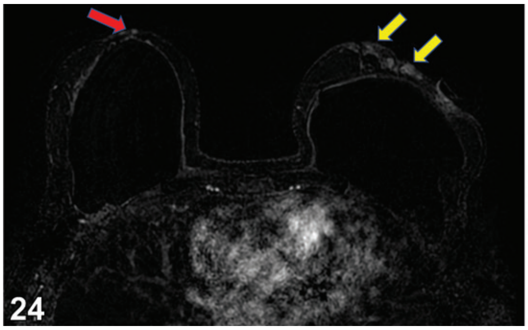
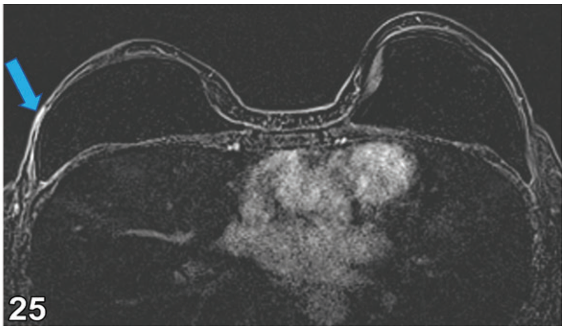



**Figure 23.** Fat necrosis in a 63-year-old woman 1 year after bilateral flap reconstruction. Fluorodeoxyglucose-avid soft tissue in the posteromedial reconstructed left breast was found at PET/CT (not shown). Recurrence was not excluded, and US (not shown) showed an irregular mixed-echogenicity mass. US-guided biopsy results showed fat necrosis. **(A)** Axial precontrast non–fat-saturated T1-weighted MR image shows a mass (arrow) with irregular margins and a central area of high signal intensity in the posteromedial left breast, along the zone of contact. **(B)** Axial contrast-enhanced fat-saturated T1-weighted MR image shows the mass (arrow) as fat signal intensity with a thin rim of peripheral enhancement, which is consistent with fat necrosis.



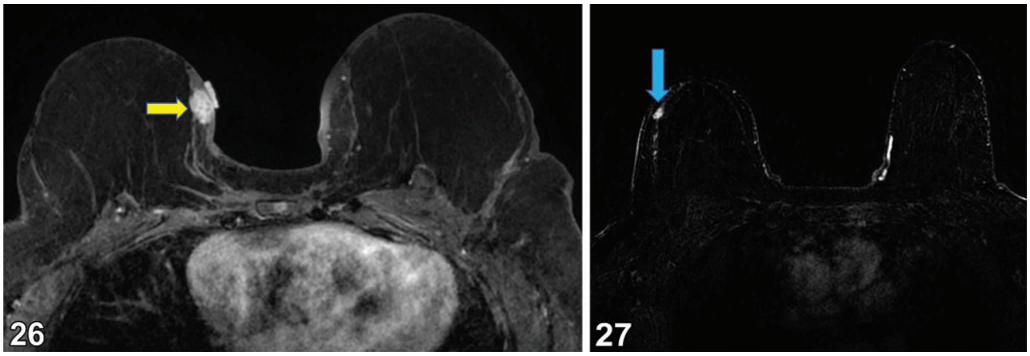

**Figures 24, 25.** **(24)** Dermal recurrence in a 55-year-old woman after NSM and implant reconstructions for bilateral breast cancer 4 years earlier. Axial subtraction MRI image shows a normal-appearing reconstructed right breast and nipple (red arrow) and abnormal nodular enhancing skin thickening of the reconstructed left breast (yellow arrows), which was palpable. Biopsy results confirmed a dermal recurrence. **(25)** Dermal recurrence in a 60-year-old woman with a history of cancer in the right breast and bilateral mastectomy, with retropectoral saline implant reconstructions. Axial subtraction MR image shows 1.9-cm linear enhancing skin thickening with washout kinetics (arrow).

The procedure is generally well tolerated, but complications can include fat necrosis, which may be palpable and can lead to diagnostic evaluation and possible intervention (64).

Use of an acellular dermal matrix has also increased in conjunction with the trend of placing implants either in a prepectoral or dual-plane location, where only the upper one-third of the implant is covered by muscle (63,65). There are several matrices available that act as a scaffold for formation of endogenous connective tissue and are gradually reabsorbed while acting as a scaffold for formation of endogenous connective tissue (65). The main advantages to use of these matrices include better control of the positioning of the implant, more coverage of the lower part of the implant, and inferior support for the implant. Although the use of an acellular dermal matrix is in its infancy, one disadvantage is an associated increased complication rate, according to Aguilera-Saez et al (65).

## Imaging Appearance of Recurrence

Most recurrences of cancer in the reconstructed breast occur 1–5 years after mastectomy, but they can occur later. Most recurrences do not occur at the nipple, but as distant metastases or regional axillary recurrences. The incidence of recurrence in the NAC itself is low, at 0%–1%, according to researchers (5,6). Recurrence in the reconstructed breast can occur in the skin or the subcutaneous tissue of the flap (Figs 24, 25), at the contact zone between the flap reconstruction and native tissue. (Figs 26, 27), or in the posterior aspect of the flap. Recurrence along the zone of contact can appear as focal enhancing thickening or enhancing masses. Dermal recurrence is not uncommon in patients who undergo NSM, especially in those who do not undergo adjuvant radiation therapy. Careful attention to new areas of enhancement in the skin, including linear or nodular enhancement and focal or diffuse enhancement, is important in evaluating for recurrent disease. Recurrence in the



**Figures 26, 27.** **(26)** Recurrence of breast cancer in a 55-year-old woman who underwent bilateral mastectomy and flap reconstruction for cancer of the right breast 2 years previously and presented with a palpable mass in the right breast. Axial contrast-enhanced fat-saturated T1-weighted MR image shows multiple enhancing masses along the zone of contact, the largest of which was palpable (arrow), with a fiducial marker placed on the skin before imaging. The palpable mass was fluorodeoxyglucose-avid at PET/CT (not shown) and did not show fat signal intensity, which would have been suggestive of fat necrosis, at axial T1-weighted non–fat-saturated T1-weighted MRI (not shown). US-guided core biopsy results of the palpable mass showed recurrent right breast adenocarcinoma, despite negative margins at the time of the patient's original surgery. **(27)** Recurrence of breast cancer in a 73-year-old woman who underwent mastectomy and flap reconstruction for breast cancer 12 years before. Axial subtraction MR image shows an enhancing mass (arrow) along the zone of contact. Targeted US and US-guided biopsy results showed recurrence of cancer.

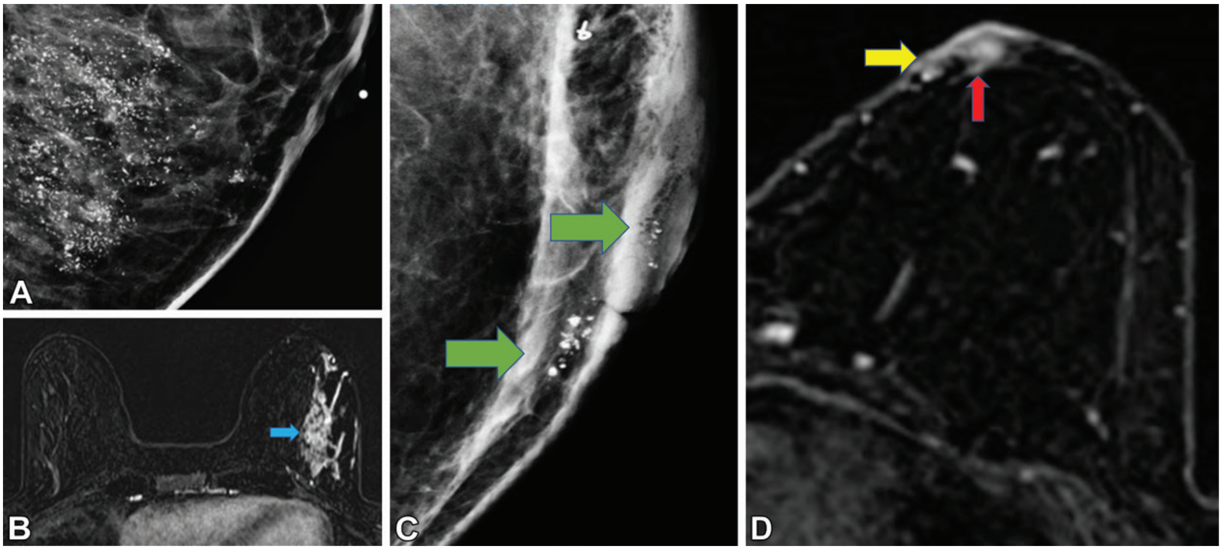


**Figure 28.** DCIS and subsequent Paget disease 2 years after NSM. **(A)** Initial mammogram shows suspicious regional calcifications in the left breast. Stereotactic biopsy results showed high-grade DCIS. **(B)** Subsequent breast MR image shows nonmass enhancement (arrow), extending close to the skin and within 2 cm of the NAC. The patient underwent NSM. **(C)** Mammogram acquired 2 years later, when the patient presented with left nipple discharge, shows heterogeneous calcifications in the nipple and retroareolar region (arrows). Biopsy results showed Paget disease. **(D)** Subsequent breast MR image shows abnormal enhancement in the superficial linear enhancement zone (yellow arrow) of the nipple and retroareolar location at the zone of contact (red arrow).

posterior aspect of the transverse rectus abdominis myocutaneous flap or the latissimus dorsi flap should be described as either at or deeper than the pedicle. Unless it is definitively benign, a new enhancing lesion at breast MRI in patients who have undergone mastectomy and breast reconstruction for cancer is concerning for recurrence, and tissue diagnosis should be obtained.

A recurrence in the flap can have a similar imaging appearance to that of the primary cancer, or it can be nonspecific (51). The mammographic manifestations of recurrent breast cancer may include a spiculated mass, distortion, microcalcifications, and skin thickening; alternatively, recurrence may mimic the appearance of fat necrosis (51) (Fig 28). Because recurrence can have a nonspecific appearance at imaging, a new palpable mass in a patient who previously underwent mastectomy and reconstruction should be treated with a high index of suspicion and a low threshold for image-guided biopsy. Recurrence is rare for DCIS that was treated with mastectomy. DCIS involves the NAC less frequently than does invasive ductal or lobular carcinoma,

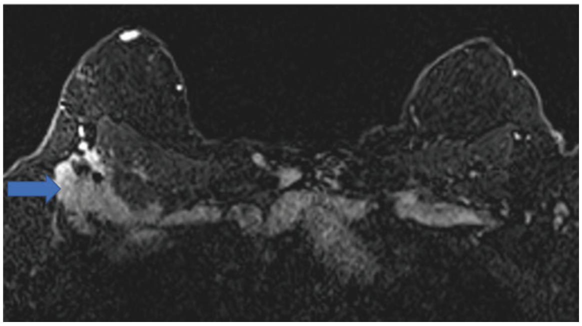


**Figure 29.** Recurrence in a 54-year-old woman with a history of infiltrating lobular cancer in the right breast and bilateral mastectomy and flap reconstruction who was found to have multiple matted lymph nodes in her right axilla at CT of the chest 8 years later. Axial contrast-enhanced fat-saturated breast MR image shows an infiltrative right axillary mass (arrow) that encases the right pectoralis minor muscle and axillary vessels and invades the right pectoralis major muscle.

but low recurrence rates are also attributed to appropriate selection criteria for NSM. Recurrence frequently manifests as a heterogeneously enhancing mass of intermediate-to-low signal intensity at T1-weighted MRI and intermediate-to-high signal intensity with rapid enhancement after administration of contrast material (54). In spite of high signal intensity at T2-weighted MRI, which simulates a fluid collection, a mass with rapid enhancement can be related to a rare but possible mucinous carcinoma. Invasive cancer may less commonly appear as a well-circumscribed mass or may simulate a postoperative fluid collection (54). Recurrence can also manifest as diffuse tumor infiltration of the flap, more commonly occurring in patients with invasive lobular carcinoma (54). When recurrence occurs in the skin, it can occur as a focal enhancing dermal lesion or a larger area of enhancing skin thickening (Figs 24, 25).

With recurrence in the posterior aspect of the flap and parasternal locations, breast MRI allows evaluation of the depth and extent of invasion and the involvement of adjacent structures including the sternum and chest wall (Fig 29). This can facilitate excision to preserve the reconstruction. PET/CT is often performed to assess for distant metastases (52).

## Conclusion

On the basis of the available literature, NSM is an increasingly common surgical treatment option. Oncologic safety is the most important consideration in patient selection, and radiologists must be aware of the normal appearance, normal variants, and abnormal appearance of the NAC. They should include the appearance of the NAC when reading preoperative imaging studies in patients with newly diagnosed malignancy and when reporting on postoperative studies of patients who have undergone NSM.

Although absolute guidelines for performing the procedure still do not exist, many surgeons use a minimum TND of 1 cm and the absence of imaging and clinical involvement of the NAC, including Paget disease and inflammatory carcinoma of the breast, as guidelines for candidate selection. All modalities of breast imaging, and breast MRI in particular, are important in providing necessary information to the treatment team to guide selection of candidates for NSM and reduce the risk of local-regional recurrence (25,44,45).

Although breast MRI of patients who have undergone mastectomy (including NSM) is not recommended for all indications, it is appropriate for some women, including those at high risk for disease who have a remaining native breast. It may be appropriate in patients with a new clinical concern after an initial imaging evaluation (49). Radiologists should be aware of the normal imaging appearance after mastectomy and NSM, that of common complications, and those that are suggestive of malignancy to avoid unnecessary additional imaging and procedures and to detect local recurrence as early as possible.

## References

1. Rosenberg K. Mastectomy rates rising in women who don't require mastectomy. Am J Nurs 2015;115(2):56–57.
2. Jonczyk MM, Jean J, Graham R, Chatterjee A. Surgical trends in breast cancer: a rise in novel operative treatment options over a 12 year analysis. Breast Cancer Res Treat 2019;173(2):267–274.
3. Li M, Chen K, Liu F, Su F, Li S, Zhu L. Nipple sparing mastectomy in breast cancer patients and long-term survival outcomes: An analysis of the SEER database. PLoS One 2017;12(8):e0183448.
4. Vlajcic Z, Zic R, Stanec S, Lambasa S, Petrovecki M, Stanec Z. Nipple-areola complex preservation: predictive factors of neoplastic nipple-areola complex invasion. Ann Plast Surg 2005;55(3):240–244.
5. Wu ZY, Kim HJ, Lee JW, et al. Breast cancer recurrence in the nipple-areola complex after nipple-sparing mastectomy with immediate breast reconstruction for invasive breast cancer. JAMA Surg 2019;154(11):1030–1037.
6. Smith BL, Tang R, Rai U, et al. Oncologic Safety of Nipple-Sparing Mastectomy in Women with Breast Cancer. J Am Coll Surg 2017;225(3):361–365.
7. Kracoff-Sella SL, Allweis TM, Bokov I, et al. tumor-to-nipple distance in selecting patients for nipple-sparing mastectomy. Plast Reconstr Surg Glob Open 2020;8(7):e2963.
8. Woodward S, Willis A, Lazar M, Berger AC, Tsangaris T. Nipple-sparing mastectomy: A review of outcomes at a single institution. Breast J. 2020 Nov;26(11):2183-2187.
9. Wu ZY, Kim HJ, Lee J, et al. Oncologic safety of nipple-sparing mastectomy in patients with breast cancer and tumor-to-nipple distance £ 1 cm: a matched cohort study. Ann Surg Oncol 2021;28(8):4284–4291.
10. De La Cruz L, Moody AM, Tappy EE, Blankenship SA, Hecht EM. Overall survival, disease-free survival, local recurrence, and nipple-areolar recurrence in the setting of nipple-sparing mastectomy: a meta-analysis and systematic review. Ann Surg Oncol 2015;22(10):3241–3249.
11. Tousimis E, Haslinger M. Overview of indications for nipple sparing mastectomy. Gland Surg 2018;7(3):288–300.

12. Djohan R, Gage E, Gatherwright J, et al. Patient satisfaction following nipple-sparing mastectomy and immediate breast reconstruction: an 8-year outcome study. Plast Reconstr Surg 2010;125(3):818–829.
13. Wei CH, Scott AM, Price AN, et al. Psychosocial and sexual well-being following nipple-sparing mastectomy and reconstruction. Breast J 2016;22(1):10–17.
14. Mosahebi A, Ramakrishnan V, Gittos M, Collier DS. Envelope mastectomy and immediate reconstruction (EMIR), improving outcome without oncological compromise. J Plast Reconstr Aesthet Surg 2006;59(10):1025–1030.
15. Mota BS, Riera R, Ricci MD, et al. Nipple- and areola-sparing mastectomy for the treatment of breast cancer. Cochrane Database Syst Rev 2016;11(11):CD008932.
16. Yueh JH, Houlihan MJ, Slavin SA, Lee BT, Pories SE, Morris DJ. Nipple-sparing mastectomy: evaluation of patient satisfaction, aesthetic results, and sensation. Ann Plast Surg 2009;62(5):586–590.
17. Spear SL, Willey SC, Feldman ED, et al. Nipple-sparing mastectomy for prophylactic and therapeutic indications. Plast Reconstr Surg 2011;128(5):1005–1014.
18. National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology: Breast Cancer. https://www.nccn.org/professionals/physician_gls/pdf/breast_blocks.pdf. Accessed June 3, 2021.
19. Galimberti V, Vicini E, Corso G, et al. Nipple-sparing and skin-sparing mastectomy: review of aims, oncological safety and contraindications. Breast 2017;34 Suppl 1(Suppl 1):S82–S84.
20. Frey JD, Salibian AA, Lee J, et al. Oncologic Trends, Outcomes, and Risk Factors for Locoregional Recurrence: An Analysis of Tumor-to-Nipple Distance and Critical Factors in Therapeutic Nipple-Sparing Mastectomy. Plast Reconstr Surg 2019;143(6):1575–1585.
21. Jakub J, Peled A, Gray R, et al. Oncologic Safety of Prophylactic Nipple-Sparing Mastectomy in a Population With BRCA Mutations: A Multi-institutional Study. JAMA Surg 2018;153(2):123–129.
22. Santoro S, Loreti A, Cavaliere F, et al. Neoadjuvant chemotherapy is not a contraindication for nipple sparing mastectomy. Breast 2015;24(5):661–666.
23. Chen C, Hayward J, Woodard G, et al. Complete breast MRI response to neoadjuvant chemotherapy and prediction of pathologic complete response. J Breast Imaging 2019;1(3):217–222.
24. Headon HL, Kasem A, Mokbel K. The oncological safety of nipple-sparing mastectomy: a systematic review of the literature with a pooled analysis of 12,358 procedures. Arch Plast Surg 2016;43(4):328–338.
25. Piato JR, de Andrade RD, Chala LF, et al. MRI to predict nipple involvement in breast cancer patients. AJR Am J Roentgenol 2016;206(5):1124–1130.
26. Veronesi U, Stafyla V, Luini A, Veronesi P. Breast cancer: from "maximum tolerable" to "minimum effective" treatment. Front Oncol 2012;2:125.
27. Luo D, Ha J, Latham B, et al. The accuracy of intraoperative subareolar frozen section in nipple-sparing mastectomies. Ochsner J 2010;10(3):188–192.
28. Sacchini V, Pinotti JA, Barros AC, et al. Nipple-sparing mastectomy for breast cancer and risk reduction: oncologic or technical problem? J Am Coll Surg 2006;203(5):704–714.
29. Nicholson BT, Harvey JA, Cohen MA. Nipple-areolar complex: normal anatomy and benign and malignant processes. RadioGraphics 2009;29(2):509–523.
30. Chiorean A, Pintican RM, Szep M, et al. Nipple ultrasound: a pictorial essay. Korean J Radiol 2020;21(8):955–966.
31. Gao Y, Brachtel EF, Hernandez O, Heller SL. An analysis of nipple enhancement at breast mri with radiologic-pathologic correlation. RadioGraphics 2019;39(1):10–27.
32. Burk KS, Seiler SJ, Porembka JH. Diagnosis, management, and percutaneous sampling of nipple-areolar calcifications: how radiologists can help patients avoid the operating room. AJR Am J Roentgenol 2021;216(1):48–56.
33. del Riego J, Pitarch M, Codina C, et al. Multimodality approach to the nipple-areolar complex: a pictorial review and diagnostic algorithm. Insights Imaging 2020;11(1):1–27.
34. Yoon JH, Yoon H, Kim EK, Moon HJ, Park YV, Kim MJ. Ultrasonographic evaluation of women with pathologic nipple discharge. Ultrasonography 2017;36(4):310–320.
35. Guo R, Lu G, Qin B, Fei B. Ultrasound Imaging Technologies for Breast Cancer Detection and Management: A Review. Ultrasound Med Biol 2018;44(1):37–70.
36. Ferré R, Paré M, Smith L, et al. Retroareolar carcinomas in breast ultrasound: pearls and pitfalls. Cancers (Basel) 2016;9(1):1.
37. Sickles EA, D'Orsi CJ, Bassett LW, et al. ACR BI-RADS Mammography. In: ACR BI-RADS Atlas, Breast Imaging Reporting and Data System. Reston, Va: American College of Radiology, 2013.
38. Chan SE, Liao CY, Wang TY, et al. The diagnostic utility of preoperative breast magnetic resonance imaging (MRI) and/or intraoperative sub-nipple biopsy in nipple-sparing mastectomy. Eur J Surg Oncol 2017;43(1):76–84.
39. Barry J, de Alcantara Filho P, D'Arcy C, Stempel M, Sacchini V. To determine the predictive value of preoperative breast MRI in selecting patients appropriate for nipple-sparing mastectomy. J Clin Oncol 2011;29(15 suppl):e11579.
40. Da Costa D, Taddese A, Cure ML, Gerson D, Poppiti R Jr, Esserman LE. Common and unusual diseases of the nipple-areolar complex. RadioGraphics 2007;27(Suppl 1):S65–S77.
41. Lim S, Park G, Choi HJ, Kwon WJ, Kang BS, Bang M. Use of preoperative mammography, ultrasonography, and MRI to predict nipple areolar complex involvement in breast cancer. Br J Radiol 2019;92(1102):20190074.
42. Fushimi A, Kudo R, Takeyama H. Do Decreased Breast Microcalcifications After Neoadjuvant Chemotherapy Predict Pathologic Complete Response? Clin Breast Cancer 2020;20(1):e82–e88.
43. Feliciano Y, Mamtani A, Morrow M, Stempel MM, Patil S, Jochelson MS. Do calcifications seen on mammography after neoadjuvant chemotherapy for breast cancer always need to be excised? Ann Surg Oncol 2017;24(6):1492–1498.
44. Sakamoto N, Tozaki M, Hoshi K, Fukuma E. Is MRI useful for the prediction of nipple involvement? Breast Cancer 2013;20(4):316–322.
45. Moon JY, Chang YW, Lee EH, Seo DY. Malignant invasion of the nipple-areolar complex of the breast: usefulness of breast MRI. AJR Am J Roentgenol 2013;201(2):448–455.
46. Schoub PK. Understanding indications and defining guidelines for breast magnetic resonance imaging. SA J Radiol 2018;22(2):1353.
47. Liao CY, Wu YT, Wu WP, et al. Role of breast magnetic resonance imaging in predicting malignant invasion of the nipple-areolar complex: Potential predictors and reliability between inter-observers. Medicine (Baltimore) 2017;96(28):e7170.
48. Cho J, Chung J, Cha ES, Lee JE, Kim JH. Can preoperative 3-T MRI predict nipple-areolar complex involvement in patients with breast cancer? Clin Imaging 2016;40(1):119–124.
49. Heller SL, Lourenco AP, Neill B, et al. ACR Appropriateness Criteria Breast Cancer Screening. https://acsearch.acr.org/docs/70910/Narrative/. Accessed March 18, 2021.
50. Devon RK, Rosen MA, Mies C, Orel SG. Breast reconstruction with a transverse rectus abdominis myocutaneous flap: spectrum of normal and abnormal MR imaging findings. RadioGraphics 2004;24(5):1287–1299.
51. Pinel-Giroux FM, El Khoury MM, Trop I, Bernier C, David J, Lalonde L. Breast reconstruction: review of surgical methods and spectrum of imaging findings. RadioGraphics 2013;33(2):435–453.
52. Adrada BE, Whitman GJ, Crosby MA, Carkaci S, Dryden MJ, Dogan BE. Multimodality imaging of the reconstructed breast. Curr Probl Diagn Radiol 2015;44(6):487–495.
53. Dialani V, Lai KC, Slanetz PJ. MR imaging of the reconstructed breast: What the radiologist needs to know. Insights Imaging 2012;3(3):201–213.
54. Peng C, Chang CB, Tso HH, Flowers CI, Hylton NM, Joe BN. MRI appearance of tumor recurrence in myocutaneous flap reconstruction after mastectomy. AJR Am J Roentgenol 2011;196(4):W471–W475.
55. Radu M, Bordea C, Noditi A, Blidaru A. Assessment of mastectomy skin flaps for immediate implant-based breast reconstruction. J Med Life 2018;11(2):137–145.

56. Margolis NE, Morley C, Lotfi P, et al. Update on imaging of the postsurgical breast. RadioGraphics 2014;34(3):642–660.
57. Ramani SK, Rastogi A, Mahajan A, Nair N, Shet T, Thakur MH. Imaging of the treated breast post breast conservation surgery/oncoplasty: Pictorial review. World J Radiol 2017;9(8):321–329.
58. Chansakul T, Lai KC, Slanetz PJ. The postconservation breast: part 1, Expected imaging findings. AJR Am J Roentgenol 2012;198(2):321–330.
59. Gart MS, Smetona JT, Hanwright PJ, et al. Autologous options for postmastectomy breast reconstruction: a comparison of outcomes based on the American College of Surgeons National Surgical Quality Improvement Program. J Am Coll Surg 2013;216(2):229–238.
60. Kroll SS, Gherardini G, Martin JE, et al. Fat necrosis in free and pedicled TRAM flaps. Plast Reconstr Surg 1998;102(5):1502–1507.
61. Baumann DP, Lin HY, Chevray PM. Perforator number predicts fat necrosis in a prospective analysis of breast reconstruction with free TRAM, DIEP, and SIEA flaps. Plast Reconstr Surg 2010;125(5):1335–1341.
62. Casey WJ 3rd, Rebecca AM, Silverman A, et al. Etiology of breast masses after autologous breast reconstruction. Ann Surg Oncol 2013;20(2):607–614.
63. Darrach H, Kraenzlin F, Khavanin N, Chopra K, Sacks JM. The role of fat grafting in prepectoral breast reconstruction. Gland Surg 2019;8(1):61–66.
64. Knackstedt RW, Gatherwright J, Ataya D, Duraes EFR, Schwarz GS. Fat grafting and the palpable breast mass in implant-based breast reconstruction: incidence and implications. Plast Reconstr Surg 2019;144(2):265–275.
65. Aguilera-Sáez J, Pau Bosacoma R, Garrido Ferrer A, Guinot Madridejos A, Barret J. Early complications in cases series in implant-based immediate breast reconstruction with a biological acellular matrix during the learning curve of this technique and using 3 different matrices: a case series of 84 breasts. Int J Surg Oncol 2018;3(1):e52.